

Replication Package for Engel, Grossmann & Ockenfels (2025)

This repository contains raw data, experimental programs, and analysis code for the paper *Integrating Machine Behavior into Human Subject Experiments: A User-friendly Toolkit and an Application to Framed Prisoner's Dilemmas* by Engel, Grossmann & Ockenfels (2025).

Overview

This replication package contains:

- Raw experimental data from GPT-3.5, GPT-4, and human participants playing repeated prisoner's dilemma games under various framing treatments
- An oTree application (`ego_human`) used to collect experimental data with human participants interacting with GPT-4
- R scripts (`plots.R` and `tables.R`) that generate all figures and tables in the paper
- Library functions (`lib/`) for data processing and statistical analysis
- Pre-generated output files (plots in PDF format and regression tables in LaTeX format)

The analysis uses R with various packages. Running both scripts (`plots.R` and `tables.R`) should take approximately 10-20 minutes on a standard desktop machine.

Data Availability and Provenance Statements

All data used in this paper were generated by the authors through experimental studies. The data were collected in two types of experiments:

1. **GPT-GPT Interactions:** The authors generated data by having GPT-3.5 and GPT-4 play repeated prisoner's dilemma games with each other under various experimental framing treatments. These interactions were conducted using the OpenAI API.
2. **GPT-Human Interactions:** Human participants were recruited online to play repeated prisoner's dilemma games with GPT-4. This data was collected using an oTree web application included in this repository.

The experimental instructions and prompts are fully documented in the `ego_human/prompts/` directory, which contains:

- System prompts for both AI (`system_c.txt`) and human participants (`system_h.txt`)
- Round-by-round instructions (`first_mover.txt`, `second_mover.txt`)
- Result feedback templates (`result.txt`)
- Three treatment variations (`prompts001`, `prompts003`, `prompts004`) corresponding to different experimental framings

Statement about Rights

- ☒ The authors of the manuscript have legitimate access to and permission to use the data used in this manuscript (the data were generated by the authors).
- ☒ The authors of the manuscript have documented permission to redistribute/publish the data contained within this replication package.

License for Data

The data are licensed under CC0 1.0 Universal (Public Domain Dedication). See `LICENSE` file for details.

Summary of Availability

- ☒ All data **are** publicly available and included in this repository.

Summary of Data Availability

Data Name	Data Files	Location	Provided
GPT-3.5 Experimental Data	<code>data35_new.csv</code>	<code>data/</code>	YES
GPT-4 Experimental Data	<code>data4_new.csv</code>	<code>data/</code>	YES
Human-GPT-4 Experimental Data (Aug 11)	<code>all_apps_wide-2023-08-11.csv</code>	<code>data/</code>	YES

Data Name	Data Files	Location	Provided
Human-GPT-4 Experimental Data (Aug 15)	all_apps_wide-2023-08-15.csv	data/	YES
Human-GPT-4 Experimental Data (Aug 16)	all_apps_wide-2023-08-16.csv	data/	YES
Human-GPT-4 Experimental Data (Aug 17)	all_apps_wide-2023-08-17.csv	data/	YES

Details on Data Files and Variables

All experimental data were collected by the authors and are provided in CSV format.

GPT-GPT Interaction Data (GPT-3.5 and GPT-4) Files: `data/data35_new.csv` (40,000 observations + header) and `data/data4_new.csv` (40,000 observations + header)

These files contain data from experiments where GPT-3.5 or GPT-4 played repeated prisoner’s dilemma games with each other under various experimental framing treatments. For each of the 10 experimental treatments, there were 200 groups of 2 GPT instances each, interacting over 10 rounds, yielding 40,000 total choice observations (10 treatments $\times$ 200 groups $\times$ 2 players $\times$ 10 rounds).

Variables:

- **experiment:** UUID identifying the experimental session
- **convo:** UUID identifying a conversation (pair of AI agents playing together)
- **thread:** UUID identifying a specific AI thread/agent instance
- **thread_type:** Type of AI thread (e.g., “GPTThread”)
- **treatment:** Integer code for experimental treatment/framing condition:
 - 1 = “base” (baseline)
 - 2 = “project”
 - 3 = “enemy”
 - 4 = “competition”
 - 5 = “war”
 - 6 = “labour”
 - 7 = “stats”
 - 8 = “couple”
 - 9 = “procurement”
 - 10 = “discretion”
- **round_:** Round number (1-10)
- **choice:** Player’s choice in this round (1 = cooperate, 2 = defect)
- **min_time:** Timestamp of the first message in the conversation (Unix timestamp)

Human-GPT-4 Interaction Data Files:

- `data/all_apps_wide-2023-08-11.csv` (50 participants + header)
- `data/all_apps_wide-2023-08-15.csv` (73 participants + header)
- `data/all_apps_wide-2023-08-16.csv` (163 participants + header)
- `data/all_apps_wide-2023-08-17.csv` (100 participants + header)

These files contain data from experiments where human participants played repeated prisoner’s dilemma games with GPT-4, conducted in August 2023 at the Cologne Laboratory for Economic Research. The raw files contain 386 total participant records. Of these, 82 participants (21%) dropped out before completing all 10 rounds and are excluded from the analysis, leaving 304 complete human-GPT-4 pairs (96 in base condition, 106 in enemy condition, 102 in competition condition). The filtering is performed in `lib/data.R` line 65 by excluding records where `ego_human.10.player.other_choice` is missing. Data is stored in wide format with separate columns for each round.

Key Variables:

- **participant.label:** Unique identifier for the human participant
- **ego_human.N.player.treatment:** Treatment/framing condition for round N (values 1, 3, or 4 corresponding to base, enemy, or competition)
- **ego_human.N.player.choice:** Human’s choice in round N (1 = cooperate, 2 = defect)

- `ego_human.N.player.other_choice`: GPT-4's choice in round N (1 = cooperate, 2 = defect)

Where N ranges from 1 to 10 representing the round number.

Dataset List

All data files are provided in CSV format (open, non-proprietary format readable by R, Python, Excel, and other software).

Data file	Source	Notes	Provided
<code>data/data35_new.csv</code>	Authors (GPT-3.5 experiments)	40,000 observations of GPT-3.5 playing prisoner's dilemma	Yes
<code>data/data4_new.csv</code>	Authors (GPT-4 experiments)	40,000 observations of GPT-4 playing prisoner's dilemma	Yes
<code>data/all_apps_wide-2023-08-11.csv</code>	Authors (human experiments)	50 participants (Aug 11 session), wide format	Yes
<code>data/all_apps_wide-2023-08-15.csv</code>	Authors (human experiments)	73 participants (Aug 15 session), wide format	Yes
<code>data/all_apps_wide-2023-08-16.csv</code>	Authors (human experiments)	163 participants (Aug 16 session), wide format	Yes
<code>data/all_apps_wide-2023-08-17.csv</code>	Authors (human experiments)	100 participants (Aug 17 session), wide format	Yes

Data integrity verification: The file `data/SHA256SUMS.asc` contains SHA-256 checksums for all data files, digitally signed with GnuPG. This allows replicators to verify that the data files have not been corrupted or modified. To verify data integrity on Linux/macOS, run:

```
cd data && gpg --quiet --decrypt SHA256SUMS.asc | sha256sum -c
```

Windows users can use equivalent tools.

Computational Requirements

Software Requirements

- ☑ The replication package contains one or more programs to install all dependencies and set up the necessary directory structure.

The analysis code requires R and several packages. The code was last run successfully with R version 4.5.0.

Required R packages (with versions used):

- `dplyr` (1.1.4): Data manipulation
- `tidyr` (1.3.1): Data reshaping (pivot functions)
- `ggplot2` (4.0.1): Plotting
- `Cairo` (1.7.0): PDF output for plots
- `patchwork` (1.3.2): Combining multiple plots
- `lmtest` (0.9.40): Testing linear regression models
- `sandwich` (3.1.1): Robust covariance matrix estimation
- `texreg` (1.39.3): Regression table formatting for LaTeX (custom fork required, see installation below)
- `plm` (2.6.7): Panel data models
- `remotes` (2.5.0): Required for installing `texreg` from GitHub

Installation:

To install all required packages automatically, run the provided installation script:

```
source("install_dependencies.R")
```

Note: The installation script installs the latest versions from CRAN, which may be newer than the versions listed above. The versions listed above are those used by the authors and are known to work correctly. For exact reproducibility, you may need to install the specific versions listed. A complete R session information file (`sessionInfo.txt`) is provided in the repository for reference.

Alternatively, you can install packages manually:

```
# Install from CRAN
install.packages(c("dplyr", "tidyr", "ggplot2", "Cairo", "patchwork",
                  "lmtest", "sandwich", "plm"))

# Install texreg from custom fork (required)
install.packages("remotes")
remotes::install_github("mrpg/texreg_fork", ref = "6268c7f")
```

Additional software (optional):

- The `ego_human/` directory contains an oTree application used to collect the human-GPT-4 experimental data. This is provided for documentation purposes and is not needed to replicate the analysis. Running this application would require:
 - Python 3.x
 - oTree package
 - The `alter_ego` package for AI integration

Controlled Randomness

- ☒ No pseudo-random number generator is used in the analysis code (`plots.R` and `tables.R`).

Note: The data generation process (GPT-GPT experiments and oTree experiments) involved randomness, but the data files provided are fixed and the analysis code processes these fixed datasets deterministically.

Memory, Runtime, Storage Requirements

Summary Time to Reproduce Approximate time needed to reproduce the analyses on a standard (2025) desktop machine:

- ☒ 10-60 minutes

Running `plots.R` takes approximately 5-15 minutes (the script processes ~86,000 rows and includes a progress counter). Running `tables.R` takes approximately 2-5 minutes.

Summary of Required Storage Space Approximate storage space needed:

- ☒ < 25 MBytes

The entire repository is approximately 12 MB (including data files of ~11 MB).

Computational Details The code was last run successfully on:

- **R version:** 4.5.0 (2025-04-11)
- **Platform:** x86_64-pc-linux-gnu
- **Operating System:** Debian GNU/Linux 13 (trixie)
- **BLAS/LAPACK:** OpenBLAS 0.3.29

The code can run on any standard desktop or laptop machine with R installed. No special hardware requirements.

Minimum requirements:

- 2 GB RAM
- Any modern CPU
- ~25 MB disk space
- Any operating system that runs R (Windows, macOS, Linux)

Description of Programs/Code

Main Analysis Scripts

plots.R

- Generates all plots/figures for the paper
- Sources helper libraries from `lib/preload.R` and `lib/data.R`
- Prints Table 3 summary statistics to console (Mean Percentage of Cooperative Choices per Platform and Frame)
- Creates the following plots in `output/`:
 - `plot1.pdf`: Cooperation conditional on platform and round (Figure 3 in paper)
 - `plot2.pdf`: Cooperation conditional on platform, round, and identity of the player (Figure 4 in paper)
 - `plotA1.pdf`: Cooperation conditional on platform, round, identity of the player and second mover’s choice in first round (Figure A1 in appendix)
 - `plotA2.pdf`: GPT-3.5 across all treatments (Figure A2 in appendix)
 - `plotA3.pdf`: GPT-4 across all treatments (Figure A3 in appendix)

tables.R

- Generates all regression tables for the paper
- Sources helper libraries from `lib/preload.R` and `lib/data.R`
- Creates the following LaTeX tables in `output/`:
 - `rt1.tex`: Framing effects in GPT-GPT interaction (Table A2)
 - `rt2.tex`: Effect of second mover defection on first mover cooperation (Table A3)
 - `rt3.tex`: Framing effects in GPT-human interaction (Table A4)
 - `rt4.tex`: Framing effects of additional frames in GPT-GPT interaction (Table A5)
 - `rt5.tex`: Effect of competitive frame in GPT-human interaction (Table A6)
 - `rt6.tex`: Cooperation by treatment and dyad composition (Table A7)

tables2.R

- Computes lower and upper bounds of cooperation rates that binomial tests reject at the 5% level
- Sources helper libraries from `lib/preload.R` and `lib/data.R`
- Defines a helper function `compute_bounds()` to systematically compute rejection bounds
- Prints results to console for Table A1 (bounds per platform, frame, and round):
 - GPT-3.5 bounds (base, enemy, competition treatments)
 - GPT-4 bounds (base, enemy, competition treatments)
 - Human bounds (base, enemy, competition treatments)
- Uses binomial tests with alternatives “greater” (for lower bounds) and “less” (for upper bounds)
- For each platform-frame-round combination, reports:
 - Lower bound: lowest cooperation rate that cannot be rejected
 - Upper bound: highest cooperation rate that cannot be rejected
 - Sample size and observed cooperation rate

Library Functions

lib/data.R

- Loads and processes all data files
- Defines key functions:
 - `augment()`: Adds timing information to conversations
 - `treatments()`: Converts treatment codes to factor labels
 - `to_long()`: Converts wide-format human data to long format
- Computes derived variables:
 - `coop`: Binary cooperation indicator (`choice == 1`)
 - `second`: Indicator for second mover in conversation
 - `ldef`: Whether opponent last defected
 - `lsdef`: Sum of opponent defections so far
- Combines all data sources into unified datasets

lib/preload.R

- Loads required packages
- Defines helper functions for regression analysis:

- `tt()`: Formats regression tables for LaTeX output
- `save_to()`: Saves multiple objects to files
- `robustly()`: Computes robust standard errors (HC3)
- `standard()`: Extracts standard errors
- `clustered()`: Computes cluster-robust standard errors
- `ci()`: Calculates confidence intervals using proportion tests

Experimental Program

`ego_human/`

- oTree application used to collect human-GPT-4 experimental data
- Contains:
 - `__init__.py`: Main oTree app logic (treatment assignment, game flow, payoff calculations)
 - HTML templates for participant interface
 - `prompts/`: Experimental instructions and prompts for different treatments
- This is provided for documentation purposes; it is not needed to replicate the analysis

License for Code

Unless otherwise specified below, all files in this repository are licensed under GNU General Public License v3.0 (GPL-3.0) due to GPL-licensed R package dependencies. See `LICENSE-CODE` for full license text.

The following files are dual-licensed under GPL-3.0 AND CC0 1.0 (you may choose either license):

- `*.csv`, `*.md`, `*.txt`, `*.pdf`, `*.tex`, `*.asc`
- `ego_human/*.py`, `ego_human/*.html`

See `LICENSE` (CC0) and `LICENSE-CODE` (GPL-3.0) for full legal texts.

Instructions to Replicators

Step-by-Step Instructions

- 1. Install R and required packages:**
 - Install R (version 4.x or later recommended)
 - Run the installation script to install all required packages:


```
source("install_dependencies.R")
```
 - Alternatively, see “Software Requirements” section above for manual installation commands
- 2. Clone or download this repository:**
 - All data files are included in the `data/` directory
 - No additional data download is necessary
- 3. Set working directory:**
 - In R, set your working directory to the root of this repository:


```
setwd("/path/to/ego_paper_data")
```
- 4. Generate all figures:**
 - Run the plots script:


```
source("plots.R")
```
 - This will take 5-15 minutes and create PDF files in the `output/` directory
 - Progress will be displayed as the script processes approximately 86,000 rows
- 5. Generate all tables:**
 - Run the main tables script:


```
source("tables.R")
```
 - This will take 2-5 minutes and create LaTeX table files in the `output/` directory
 - Optionally, run the bounds analysis script:


```
source("tables2.R")
```
 - This will take 1-2 minutes and print Table A1 (cooperation rate bounds) to the console
- 6. Check output:**
 - All figures (PDF format) should appear in `output/`: `plot1.pdf`, `plot2.pdf`, `plotA1.pdf`, `plotA2.pdf`, `plotA3.pdf`
 - All tables (LaTeX format) should appear in `output/`: `rt1.tex` through `rt6.tex`
 - Table A1 bounds will be printed to the console when running `tables2.R`

Notes

- The scripts can be run in any order (plots.R, tables.R, and tables2.R are independent)
- All scripts will display progress messages during execution
- The `lib/data.R` script displays “Please wait until the counter reaches ~86000” while computing derived variables
- `plots.R` prints Table 3 summary statistics to the console during execution
- `tables2.R` prints Table A1 bounds analysis to the console
- Pre-computed output files are already included in the `output/` directory for comparison

List of Tables and Programs

The provided code reproduces:

- ☒ All tables and figures in the paper

Figures and Summary Statistics

Figure/Table #	Program	Line Range	Output File	Notes
Table 3	plots.R	18-26	Console output	Mean percentage of cooperative choices per platform and frame
Figure 3	plots.R	28-37	output/plot1.pdf	Cooperation conditional on platform and round
Figure 4	plots.R	42-94	output/plot2.pdf	Cooperation conditional on platform, round, and identity of player
Figure A1	plots.R	96-161	output/plotA1.pdf	Cooperation conditional on platform, round, identity and initial choice
Figure A2	plots.R	164-181	output/plotA2.pdf	GPT-3.5 across all treatments
Figure A3	plots.R	184-201	output/plotA3.pdf	GPT-4 across all treatments

Tables

Table #	Program	Line Range	Output File	Notes
Table A1	tables2.R	Full script	Console output	Lower and upper bounds of cooperation rates per platform, frame, and round

Table #	Program	Line Range	Output File	Notes
Table A2	tables.R	6-40	output/rt1.tex	Framing effect in GPT-GPT interaction
Table A3	tables.R	43-85	output/rt2.tex	Effect of second mover defection on first mover cooperation
Table A4	tables.R	87-118	output/rt3.tex	Framing effect in GPT-human interaction
Table A5	tables.R	120-156	output/rt4.tex	Framing effects of additional frames in GPT-GPT interaction
Table A6	tables.R	159-185	output/rt5.tex	Effect of competitive frame in GPT-human interaction
Table A7	tables.R	187-217	output/rt6.tex	Cooperation by treatment and dyad composition

References

No external data sources or references are cited in this README. All data were generated by the authors as described above.

Repository Structure

```
ego_paper_data/
  README.md          # This file
  LICENSE            # CC0 1.0 Universal license (for data/documentation)
  LICENSE-CODE       # GPL-3.0 license (for R code)
  install_dependencies.R # Script to install all required R packages
  sessionInfo.txt    # R session information with package versions
  data/              # Raw experimental data (CSV format)
    data35_new.csv   # GPT-3.5 experimental data
    data4_new.csv    # GPT-4 experimental data
    all_apps_wide-2023-08-11.csv # Human-GPT-4 data (session 1)
    all_apps_wide-2023-08-15.csv # Human-GPT-4 data (session 2)
    all_apps_wide-2023-08-16.csv # Human-GPT-4 data (session 3)
    all_apps_wide-2023-08-17.csv # Human-GPT-4 data (session 4)
    SHA256SUMS.asc   # Checksums for data files
  lib/               # Library functions for analysis
    data.R           # Data loading and processing
    preload.R        # Helper functions and package loading
  plots.R            # Script to generate all figures
  tables.R           # Script to generate all tables
  tables2.R          # Script to compute cooperation rate bounds (Table A1)
  output/            # Generated output files
    plot1.pdf        # Figure 3
    plot2.pdf        # Figure 4
```


plotA1.pdf	# Appendix Figure A1
plotA2.pdf	# Appendix Figure A2
plotA3.pdf	# Appendix Figure A3
rt1.tex	# Table A2
rt2.tex	# Table A3
rt3.tex	# Table A4
rt4.tex	# Table A5
rt5.tex	# Table A6
rt6.tex	# Table A7
ego_human/	# oTree experimental application (documentation)
__init__.py	# Main oTree app logic
*.html	# Participant interface templates
prompts/	# Experimental instructions and prompts
prompts001/	# Treatment 1 (base)
prompts003/	# Treatment 3 (enemy)
prompts004/	# Treatment 4 (competition)

Acknowledgements

This README template is based on the template provided by the Social Science Data Editors, adapted for this specific replication package.